

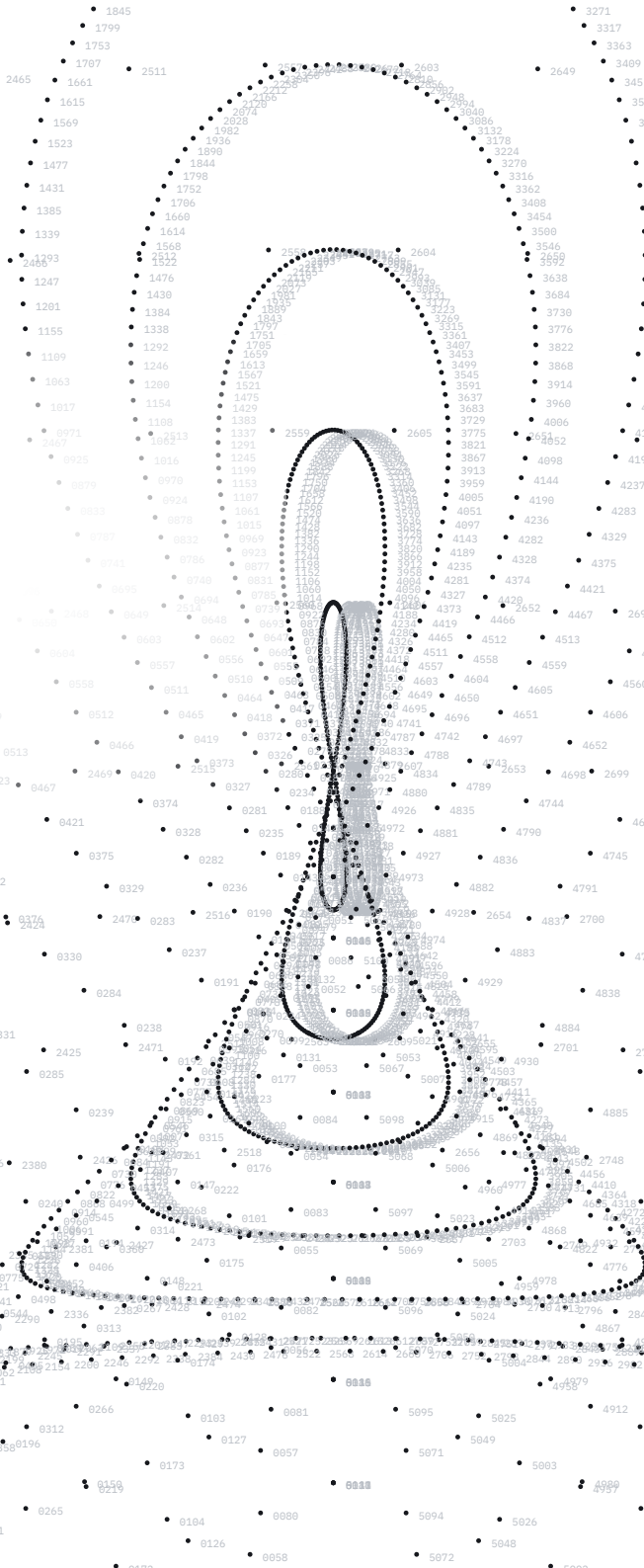
SELECTED WORK · 2023-2026

Charles Abi Chahine.

architect · computational designer

design, computation, and the work of getting it built.

charlesabichahine.com



				
01	03	05	07	08

01		Sensi	A sensory copilot for floor plans.	2026
02		Encoding Urban Risk	Street risk classified from open map data.	2026
03		IEgoarCh	A prompt becomes a buildable LEGO set.	2026
04		Breathing Mass	A tower that breathes for Santiago.	2026
05		The Huddle	A research hub huddled against the wind.	2025/26
06		The Luminous Stratum	Sedimented light over a Cairo market.	2025
07		Rings of Mars: Ring 4000	A regolith-printed habitat ring for Mars.	2024
08		Saria, Dubai Maritime City	A residential tower on the water in Dubai.	2024



Sensi.

2026

COMPUTATION & AI

MACAD 2026 AWARD

A sensory copilot for floor plans.

In architecture we model everything: structure, cost, energy, code compliance. Layer after layer that makes a building accountable before it is built. Sensi adds the one layer we never formalized: how a space will actually feel. Not emotion, but the full sensory experience of standing in a room, scored across six senses: thermal, visual, acoustic, spatial, olfactory, tactile. Not in the abstract, but for a specific person.

MODULE

AIA Studio · MaCAD, IAAC

TEAM

Charles Abi Chahine, Emilie El Chidiac,
Lakzhmy Mari Zaro, María Sánchez
Domínguez

TOOLS

Python · LLM agents · Vision models ·
Web app



01 · THE SENSORY LAYER

Comfort research studies one sense at a time, but we take a room in through all of them at once, and they are coupled: one moderates another. Add a bigger window and the daylight improves, but the same glass is a thinner sound barrier and leaks heat. The room score is half mean, half worst, because the worst sense is the one you would actually feel. Scoring is deterministic; the LLM routes intent and explains results, it never decides them.

02 · ACT 1 · ONBOARD: WHO IS THIS FOR

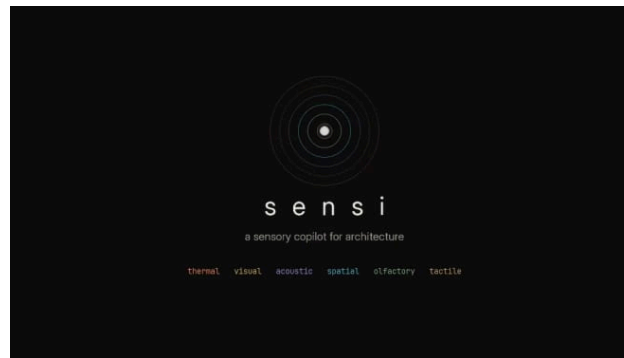
Comfort is not universal, so Sensi opens by learning who you are: a few questions, then a moodboard where each image you keep quietly tags the senses you lean toward. It compiles into a persona drawn as a petal rose.

03 · ACT 2 · SHAPE: COMFORT YOU CAN EDIT

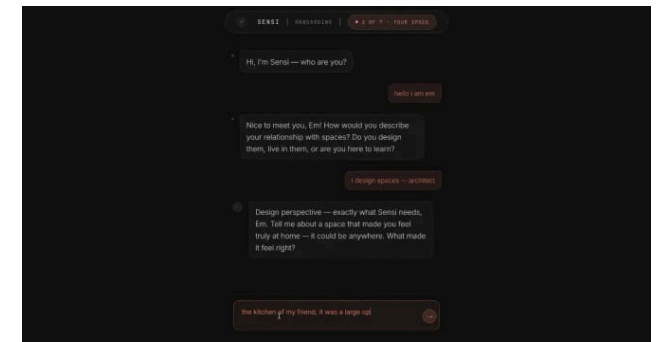
The heart of the system. You talk to it in plain language and a fast routing model classifies your intent in about 0.6 seconds. An edit is planned, validated against the plan, applied and re-scored live, and kept as a checkpoint. Rooms are nodes and doors are edges, so comfort becomes a zoning problem you can see.

04 · ACT 3 · REPORT: CLOSING THE LOOP

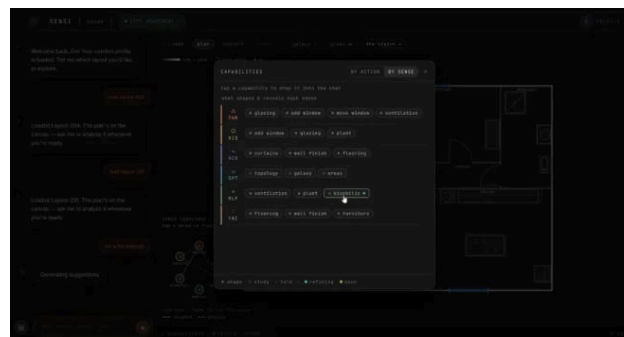
Final scores write a prompt and a vision model renders each room under an honest rule: only the extreme senses speak, the comfortable middle stays quiet. You compare it back to the moodboard from Act 1. About \$0.039 per room, 2.75× faster than the alternative we tested.



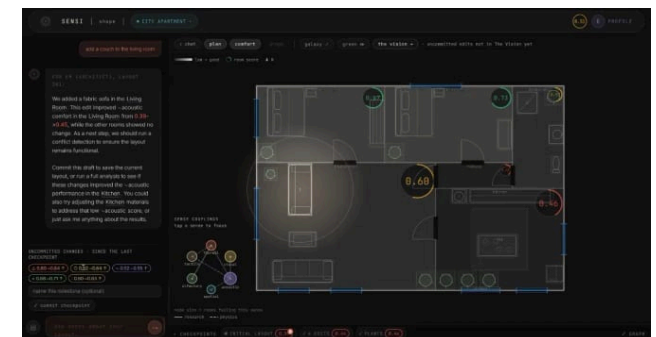
DEMO · Sensi, end to end: a plan goes in, a room-by-room comfort report comes out.



DEMO · Onboarding: from a few questions and a moodboard to a personal weighting of the six senses.



DEMO · Shaping: conversational edits that re-score the plan in real time.



DEMO · Generating the report: final scores become a grounded render of each room.



Encoding Urban Risk.

2026

COMPUTATION & AI

Street risk classified from open map data.

Can the physical layout of a street, measurable from public map data, predict how risky it feels? We framed it as a three-class problem: low, medium, or high risk, using seven spatial features drawn from OpenStreetMap and Mapillary per street segment. This is the full arc of the project, not just the results, but the wrong turn that made them honest.



FIG 01

SHAP: London's high-risk drivers stay in-distribution; the Eixample's do not.

MODULE

Data Encoding · MaCAD, IAAC

TEAM

Charles Abi Chahine, Emilie El Chidiac,
Lakzhmy Mari Zaro, María Sánchez
Domínguez

TOOLS

Python · scikit-learn · OpenStreetMap ·
Mapillary · SHAP

01 · GROUNDED IN SIXTY YEARS OF THEORY

Every feature traces to established urban-safety theory: Jacobs on eyes on the street, Newman on territorial clarity, Hillier and Hanson on network configuration. The unit is the segment, where risk clusters.

02 · HITTING THE WALL

We first tried to predict crime directly, regressing 920,000 London incidents against our features. Three models returned R^2 under 0.064: street geometry cannot capture social forces. A finding, not a failure.

03 · THE PIVOT: A SPATIAL TYPOLOGY

Rebuilt in eight steps: fetch from OSM and Mapillary, seven normalised features per segment, collapsed into a weighted score, cut into three classes with K-Means, then trained. 31% low, 47% medium, 22% high.

04 · COHERENT TYPES, AMBIGUOUS RISK

The Kohonen map organises into legible street types, but coloured by risk, near-identical cells differ sharply. Typologies describe character; they do not determine risk. That gap is the honest heart of it.

05 · WHAT ACTUALLY MATTERS

Logistic Regression hits 99% and Random Forest 95%, expected since the label came from the same features. The ablation says more: transport leads, then connectivity and land use. Lighting barely registers.

06 · LONDON TRAINED, WORLD TESTED

Applied to Barcelona's Eixample and Trastevere, the reading breaks where morphology diverges. SHAP pinpoints it: land use and visibility hit values never seen. A distribution problem, not an architecture flaw.



1961 · Jacobs
ALIVE TO BE SAFE

Eyes on the street, active frontages, mixed uses & foot traffic create natural surveillance.



1972 · Newman
OWNED TO BE SAFE

Territorial control, restricted access. Small recognisable communities self-police.



1984 · Hillier & Hanson
LAYOUT DETERMINES LIFE

Street configuration shapes movement, encounter and territorial control — Space Syntax.



1984 · Brantingham
FAMILIARITY BREEDS CRIME

Offenders strike within their awareness space — familiar routes and nodes.



2008 · Hillier & Sahbaz
MORE NEIGHBOURS = SAFER

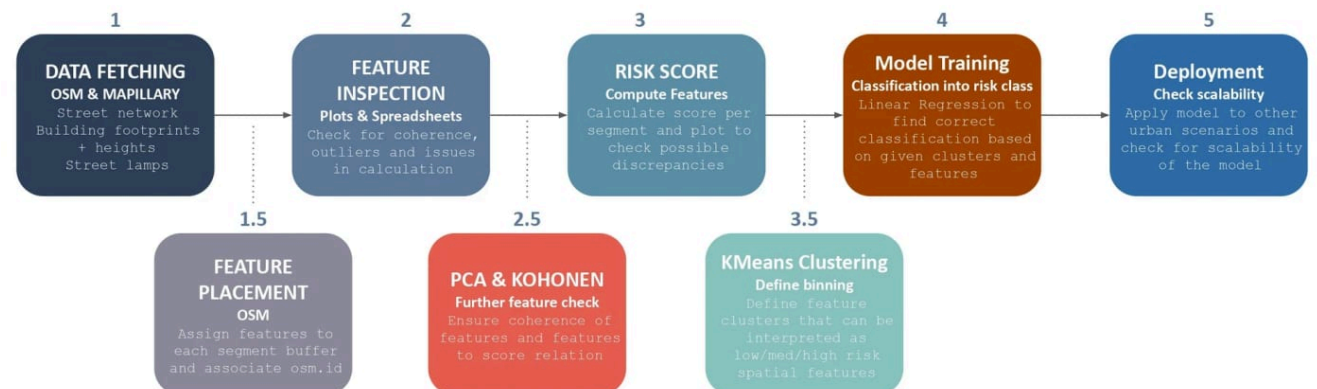
Residential density neutralises risk. Safety in numbers confirmed across 65,000+ dwellings.



2017 · Summers & Johnson
ARTERIES & SHADOWS

Global Choice is the strongest single predictor. Crime spills within 100 m of major routes.

Seven features, each anchored in a lineage of urban-safety theory.



The wall: every model flat against the diagonal. Crime does not reduce to spatial features.



LEgoarCh.

2026

COMPUTATION & AI

IAAC EXHIBITION

A prompt becomes a buildable
LEGO set.

You type a building, and a minute later a real LEGO set is sitting on a shelf: rendered, modelled, brick-built, priced, and catalog-legal. Most "AI makes LEGO" demos stop at a gorgeous render, but a render is a promise, not a product. LEgoarCh builds the downstream half: the machinery that forces the dream to obey real bricks, real colours, and real gravity.



AI PROPOSES PROBABILISTIC

01



Prompt

a building name

02



FLUX.2 + LoRA

render · 28 steps

03



TRELLIS-2

image to 3D mesh

COMPUTATION PROVES DETERMINISTIC

04



Voxelize

32 studs wide

05



Legolize

split + merge

06



Buildable set

stable & legal

MODULE

Generative AI Seminar · MaCAD, IAAC

TEAM

Charles Abi Chahine, Emilie El Chidiac

TOOLS

FLUX.2 + LoRA · TRELLIS-2 · ComfyUI · Python

01 · ONE SENTENCE, FIVE MOVES

Text prompt, FLUX render, TRELIS mesh, voxelize, legolize, a priced buildable set. The front half is probabilistic and invents the form. The back half is deterministic and proves it stands. The handoff in the middle is the entire idea.

02 · A LORA THAT SPEAKS LEGO

Base FLUX.2 does not speak fluent LEGO Architecture. We trained a LoRA on the visual grammar of real LEGO Architecture sets and swept its strength from 0 to 1.5: at 0 a plain building, at 1.0 the studs and brick seams snap in. The LEGO-ness lives in the fine-tune, not in the words.

03 · FROM ONE PHOTO TO A WHOLE OBJECT

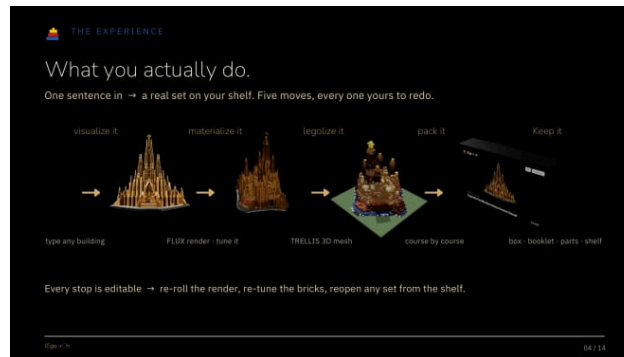
TRELIS-2 completes the full 3D form from the single render, openly guessing the unseen back. Then the deterministic half begins: the mesh is voxelized onto plate-height layers, pre-stretched 2.5× vertically because bricks are not cubes, and colours are exposure-matched back to the render.

04 · THE LEGOLIZER: WHERE BUILDABLE GETS EARNED

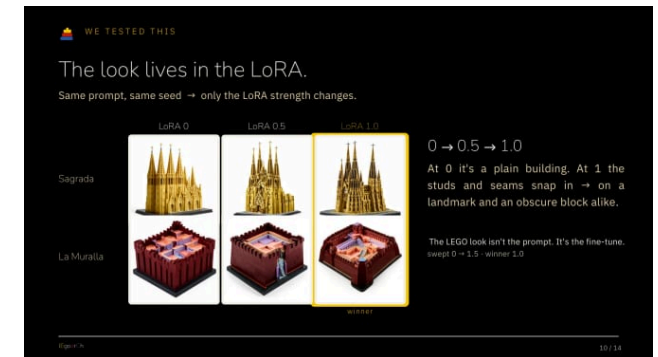
The computational centrepiece contains no AI at all. A split-and-merge solver covers each layer with the largest legal bricks that fit, with a slope pass that bevels staircases and a running-bond penalty that offsets seams. Colours snap to the nearest real LEGO colour in CIEDE2000, and 44 parts by 48 colours give 1,598 validated combinations.

05 · A SET YOU CAN KEEP

The output is a complete product: box art, a build booklet, a priced parts list, and a shelf that keeps every set you have generated.



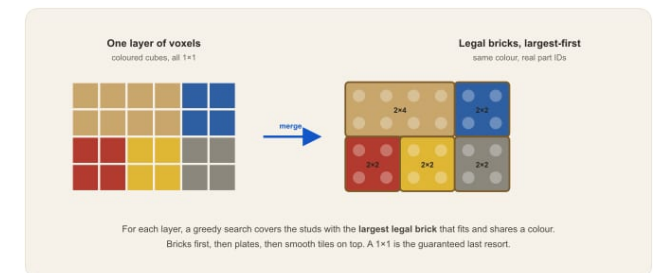
The five-move experience: visualize, materialize, legolize, pack, keep.



Same prompt, same seed, only the LoRA strength changes. 1.0 wins.



The TRELIS mesh: a couple hundred thousand triangles with a wrapped texture.



Split and merge, one layer: the biggest real bricks that share a colour, not a sea of 1×1s.



Breathing Mass.

2026

BIM & WORKFLOWS

A tower that breathes for Santiago.

Breathing Mass is a vertical ecosystem in Santiago where architecture, wind, and energy converge. The Hyper Lung captures, cleans, and redistributes polluted air through a breathing core, turning the tower into living infrastructure that breathes with the city. Within a larger hyperbuilding studio, our team owned its structure and facade: translating the lung analogy into a self-braced skeleton and an adaptive skin, every form justified by data and published to the program and data teams through Speckle.

Breathing Pattern: ONE Geometry, TWO Logics

The facade functions as a woven environmental exoskeleton where it regulates Solar radiation, Wind pressure and Air quality. So it works as an climatic filter.

Adaptive Facade

Urban Farming, Productive programs or when the interior function requires climate responsiveness.

Adaptive Facade

Residential floors, Offices, Hotels, mixed use spaces and where there is no need for daily micro-adjustments.

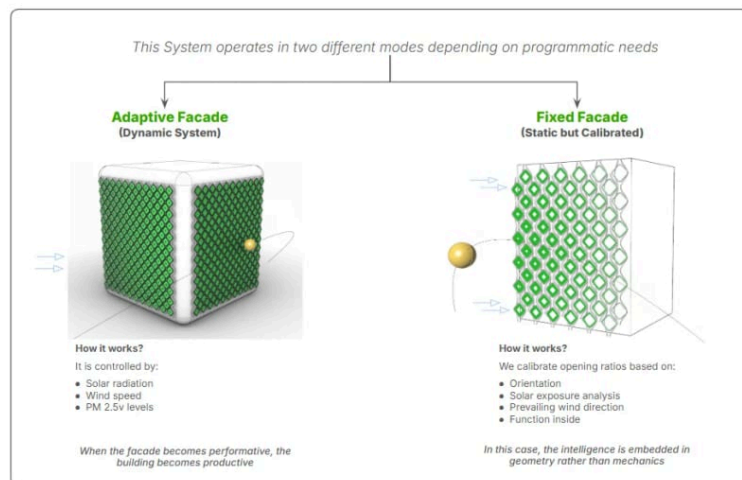


FIG 01

One grammar, two modes: an adaptive dynamic skin and a fixed collated one.

MODULE

BIMSC Studio · MaCAD, IAAC

TEAM

Charles Abi Chahine, Ramy Ayoub, Hani Karime

TOOLS

Rhino · Grasshopper · Alpaca · Speckle

01 · THE ALVEOLAR SPINE: A LUNG ANALOGY

The core takes its logic from the human lung: a porous structural core mirroring the bronchi, carrying load and moving air at once. The organising logic for both the structure and the way the tower breathes.

02 · A SELF-BRACED SKELETON

Volumes plug onto the core, and three cores form a self-bracing triangle. Alpaca read the stress, the optimization stripped what was idle, and the voxels that survived became a lattice: 1,650 tonnes down to 235.

03 · GROWING THE VOLUMES

The plugin masses are grown along the lattice, placed by incident radiation, wind pressure and volumetric density, each candidate measured across 232 branches.

04 · THE BREATHING CORE: AN ENVIRONMENTAL MACHINE

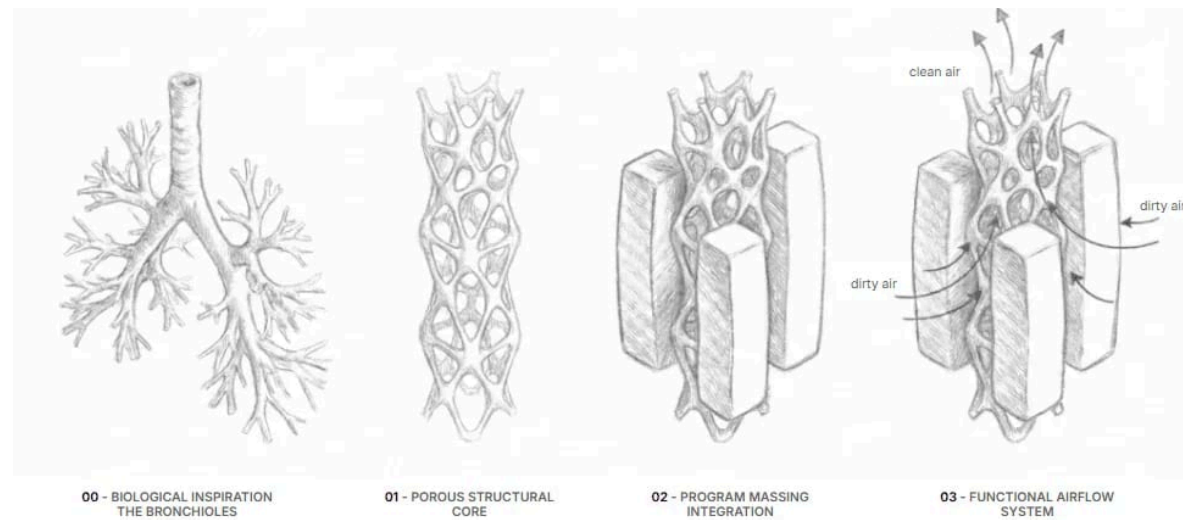
A water cascade, electrostatic precipitation and Climeworks-style capture pull CO₂ and particulates from the air before the spine redistributes it. Structure and environmental system are one object.

05 · FIXED VS. ADAPTIVE FACADE

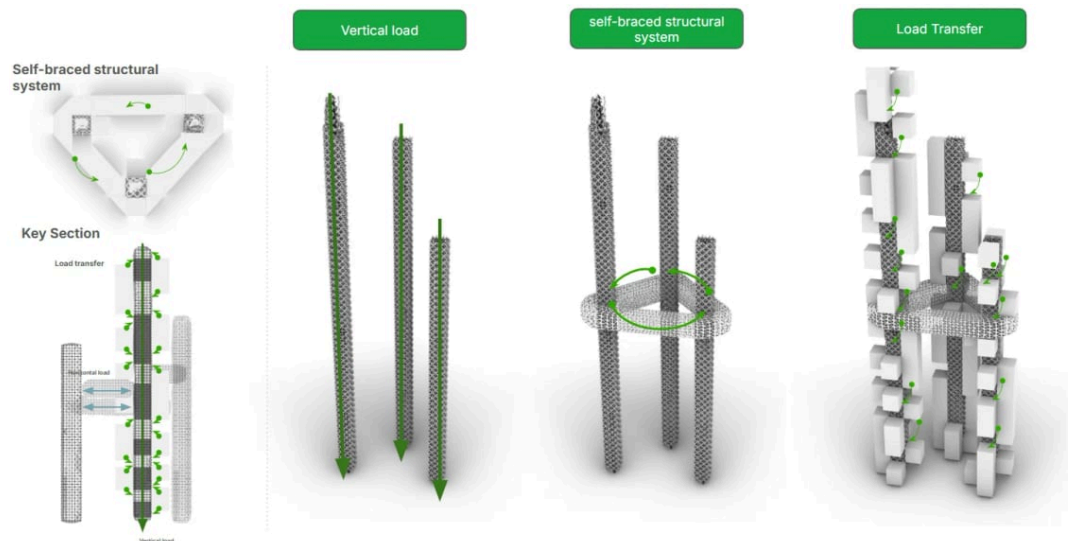
A pattern from mashrabiya logic becomes a performative facade: adaptive where the skin must respond, fixed where it need not. Wind and radiation set it, and one grammar reads both ways.

06 · THE TOWER

Assembled, it reads as plugged volumes around a breathing spine: less an object than a piece of the city's respiratory infrastructure.



From bronchi to a porous structural core: the lung analogy driving the spine.



Self-braced structural system: load transfer from plugged volumes through the triangulated cores.



The Huddle.

2025/26

DESIGN & RESEARCH

IAAC EXHIBITION

A research hub huddled against the wind.

Punta Arenas sits at the southern tip of Chile in a cold, perpetually windy subpolar climate. The Huddle is a research and education hub with an expedition basecamp for 1,000 users, designed the way penguins survive the same latitudes: compact forms, packed close, shielding one another from the wind.

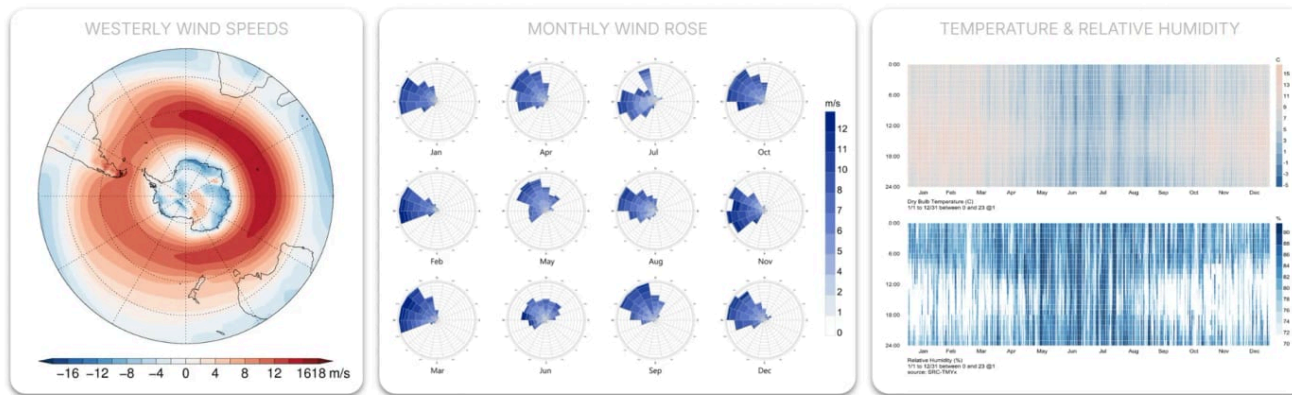


FIG 01

Westerly wind speeds across the southern hemisphere, the monthly wind rose, and a year of dry-bulb temperature and relative humidity.

MODULE

ACESD Studio · MacAD, IAAC

TEAM

Charles Abi Chahine, Emilie El Chidiac,
María Sánchez Domínguez, Lakzhmy Mari
Zaro

TOOLS

Rhino · Grasshopper · Wasp · Kangaroo ·
Speckle

01 · ENDURANCE, NOT SHELTER

Punta Arenas sits on the Magellan Strait, a departure point for Antarctic expeditions. We took the harder of two plots: open, windy, salt-worked, but facing the sea. The project accepts wind, not resists it.

02 · LEARNING FROM THE KAWI

The typology comes from the kawis, the region's native huts, already adapted to extreme wind. We reinterpreted it as 4×4×4 m units under a 16 m limit, the gaps becoming wind-protected courtyards.

03 · A WORKFLOW, NOT A SHAPE

One chain: a program definition feeds two Kangaroo solvers, then Wasp aggregates the kit of parts. Because it is a system, the part count is known at every moment and the rules hold at 50 users or 1,000.

04 · PROGRAM BECOMES MASSING

Public programme wraps around and shields the research and residential areas, on a single circulation spine. In section it stays compact under the height limit, the skin a buffer against the cold.

05 · TIMBER, SEISMIC FORCES, AND A SKIN OF THREE PANELS

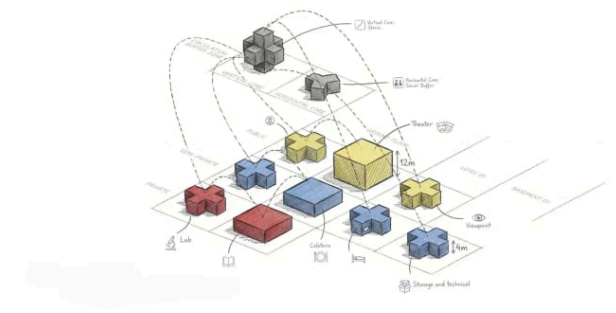
Local timber, shear walls and bracing for wind, seismic isolators for movement, stable at roughly 709 tons. Three ETFE panels: the Shield for pressure, the Lens for solar gain, the Gill for ventilation.

06 · WIND BECOMES A RESOURCE

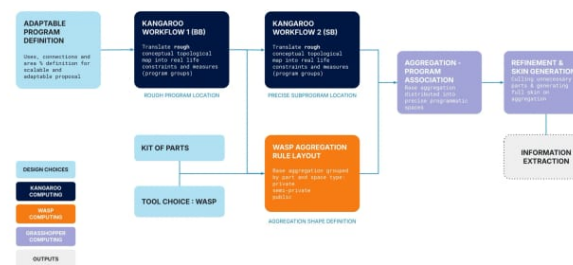
The tunnels the aggregation opens accelerate the flow, and vertical turbines stand in those zones. What the building was shaped to survive ends up powering it.



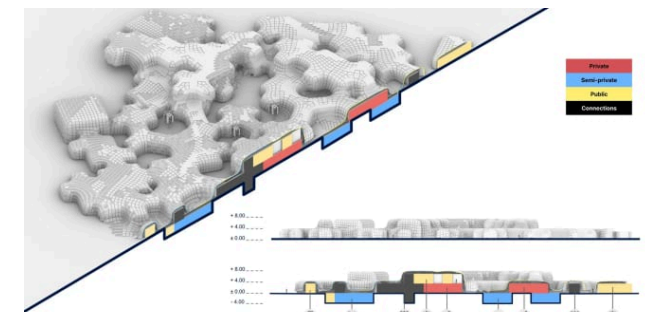
The hub on its seafront plot, the edge of Punta Arenas behind it.



The kit of parts across three levels: cross modules for general functions, reinforced cores for circulation, triangular connectors, and the 12 m theatre...



The whole definition: program definition, two Kangaroo passes, the Wasp aggregation rule layout, skin generation, information extraction.



The aggregation cut open, with two sections through it: public in yellow, semi-private in blue, private in red, connections in black.



The Luminous Stratum.

2025

DESIGN & RESEARCH

Sedimented light over a Cairo market.

In the dense historic fabric of Cairo, light is both a blessing and a burden. The Luminous Stratum proposes a new architectural language that negotiates this relationship: a “volume of sedimented light” that mimics the city’s stratification while filtering the harsh sun. Designed for the Complex Forming seminar, it is an independent system of stacked lamellas that hovers within the void of the Bab Al-Luk historic market, leaving the original structure untouched.



FIG 01

Sedimented light in the market below.

MODULE
Complex Forming · MacAD, IAAC

TEAM
Charles Abi Chahine

TOOLS
Rhino · Grasshopper · Kangaroo

01 · MORE THAN A ROOF

The system is deliberately independent: it revitalizes the market below without touching the historic walls. Voids are cut into the geometry precisely so the new form pulls away from its context. It reads as a continuous volume nested within the void, a porous buffer that protects the atrium without sealing it off.

02 · COMPUTATIONAL FORM-FINDING

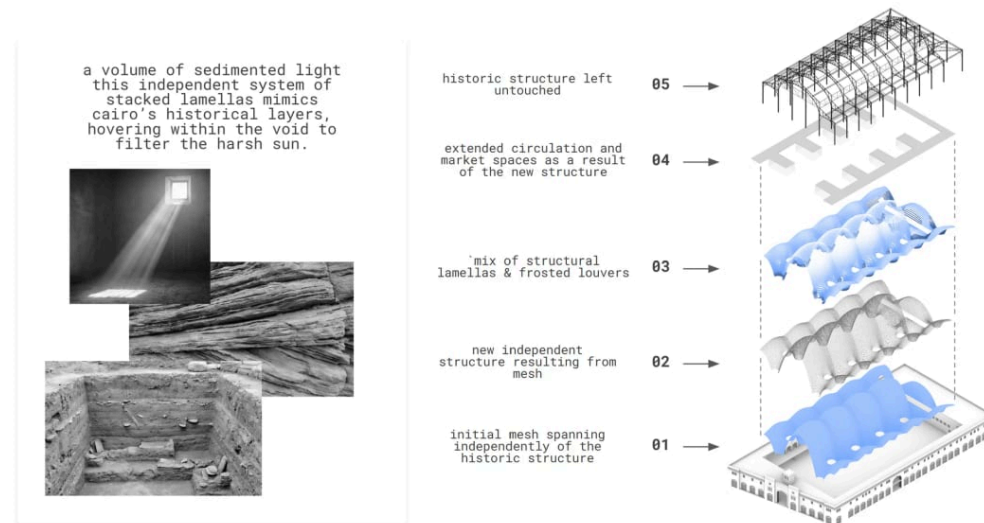
The geometry is generated through Kangaroo physics. A flat mesh, strategic voids, edge anchor points and OnCurve goals define the footprint, then vertical load forces pull the mesh into a relaxed catenary vault. The defining move comes next: mesh faces are sorted by their normal vectors, and that orientation dictates function.

03 · GEOMETRY BECOMES FUNCTION

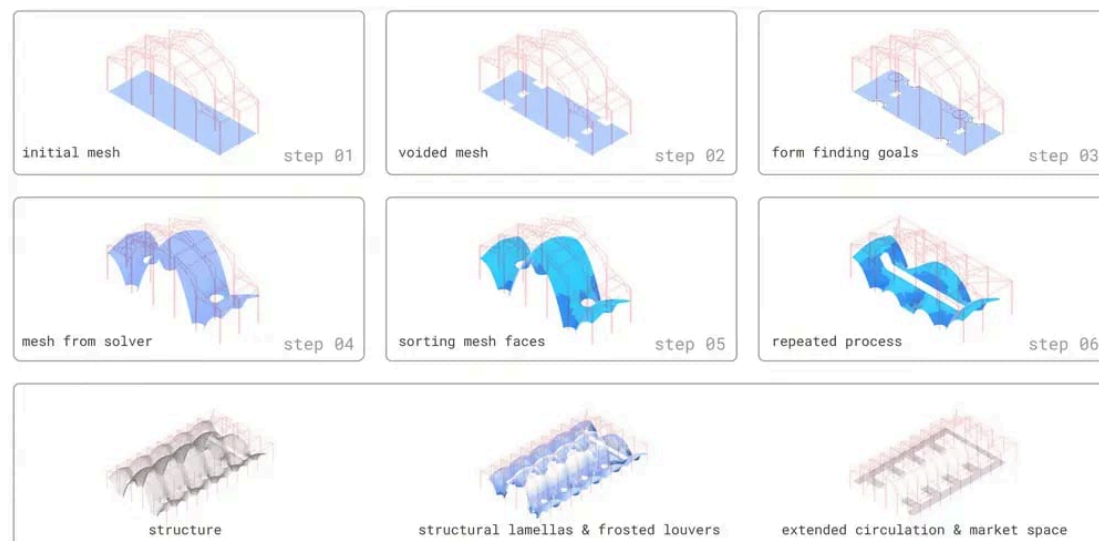
Vertical faces become gradient frosted louvers that control sun glare; horizontal faces become structural lamellas serving as shelving and walkable surfaces. All of it materializes at once from the relaxed mesh. The system is modular, and for the last vault different anchor points turn it into a continuous staircase.

04 · AN OCCUPIABLE LATTICE

I tested vertical loads from a shallow 15% to a steep 75%, and calibrated the domains for the frosted louvers and structural lamellas to tune porosity against structure. The selected iteration uses a 35% vertical load, giving the optimal vault height and clearance for the layers below.



A volume of sedimented light: an independent system layered above the untouched historic market.



DEMO · Step by step: flat mesh, strategic voids, form-finding goals, solver relaxation, face sorting.



charlesabichahine.com/work/marception



Rings of Mars: Ring 4000.

2024

DESIGN & RESEARCH

TOP 50 • VOLUME ZERO

A regolith-printed habitat ring for Mars.

How do you design a habitat for Mars? Marsception, run by Volume Zero, asked that in 2024, and our answer was a ring. Ring 4000 sits on the rugged rim of a crater inside Valles Marineris, a 3D-printed regolith shell holding farming, research, living and a transportation ring in one closed loop, with the long game that every crater gets its own ring until the rings make a community. It was an early bet that generative tools belonged on an architect's desk, made when that still raised eyebrows.

MODULE
International Competition

TEAM
Charles Abi Chahine, Emilie El Chidiac

TOOLS
Rhino • SubD • AI workflows • V-Ray

01 · A RING IN A CRATER

Three concentric circles. The inner one is a controlled farm under a glass facade, visible from every room. The middle circle is transportation. The outer is living and working space, with five sleeping pods on the lowest level. The shell is 3D-printed regolith outside and high-density polyethylene inside for an air-tight seal, with BIPV panels over the facade; the canyon walls do the rest of the shielding.

02 · WHY A RING

The craters are the site Mars already gives you, so the ring takes their shape instead of fighting it. The ecosystem closes on itself: water split into oxygen and hydrogen, plants turning the crew's CO₂ into food, the carbon and water reacting back into HDPE for printing and fuel. Transport runs in a loop, and expansion is simply the next crater over.

03 · HOW IT WAS MADE

Regolith melted into a paste and printed by drone robotics, every surface shaped to stand at an angle that needs no support. Then the model in Rhino with SubD, from the massing of the ring down to the sleeping pods. Then the part that was new at the time: feeding the geometry to generative image tools when image generation was only just becoming a thing, and finishing in V-Ray.

04 · WHAT CAME OF IT

Top 50 at Marsception, with both our names on the public results page. And our first proof that generative tools and architecture could share a desk: an early bet, and it aged well.



Ring 4000 on its crater rim inside Valles Marineris.



The competition board: the ring exploded by level, the plan, the closed loops of water, oxygen, food, HDPE and fuel, the pods and the transportation ring, and the long section through the crater pit.



charlesabichahine.com/work/saria



Saria, Dubai Maritime City.

2024

PRACTICE

A residential tower on the water in Dubai.

Saria is a residential tower on the water at Dubai Maritime City, with the Dubai skyline behind it. I took it from pre-concept through to design development as a Design Architect at SOMA: parametric facade iterations in Rhino and Grasshopper, then the BIM model for delivery. The drawings here are from the concept design submission of March 2024, with SOMA as design architect.

MODULE

SOMA · Design Architect

TEAM

SOMA: office project

TOOLS

Rhino · Grasshopper · Revit

01 · A TOWER ON THE WATER

The plot sits at the edge of the Maritime City peninsula, so the tower is read from the water first: a slender square plan rising off a podium, with the skyline of Business Bay behind it. Four podium levels carry the parking under a landscaped deck; above sit 38 levels, the roof at 167 m. The rhythm breaks once at level 20 and again at the top two floors.

02 · THE FACADE

The facade is where the parametric work went. The balconies shift and step across the elevation rather than stacking in a straight line, and the pattern was iterated in Rhino and Grasshopper: how far each band projects, where the planters sit, how the rhythm breaks. What survived went into the Revit model the drawings are cut from.

03 · THE PODIUM AND THE GROUND

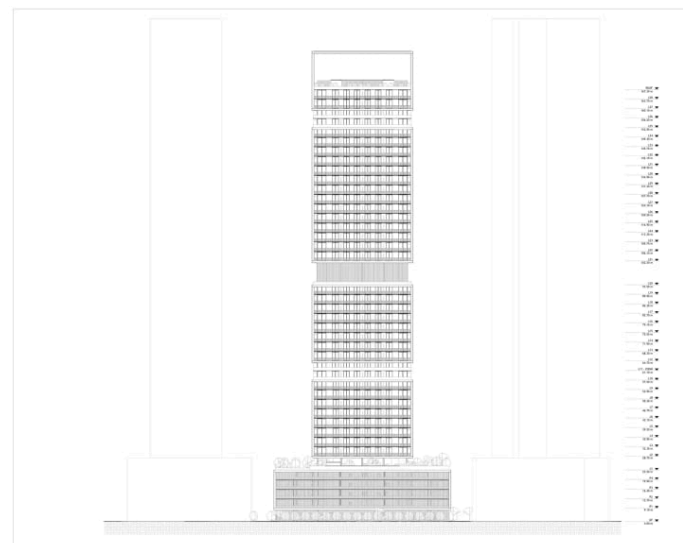
At street level the podium is wrapped in planting and opens under a deep canopy at the entrance; the landscape runs over the podium roof and down to the promenade, so the tower lands in a garden rather than on a car park.

04 · THE DRAWINGS

The concept set: the master plan on its plot, the ground floor with the lobby and drop-off, the amenity level over the podium, and a typical residential floor running from the 12th to the 34th, core in the middle and apartments around it.



Saria from the water, the podium and tower in their context of neighbouring towers.



North-west elevation: the podium, the stepping balcony bands, the break at level 20 and the crown, with the level datums from ground to the roof at 167.30 m.

Eighteen projects, four belts.

This portfolio is a selection of eight. Every project, with its full gallery, demos, and write-up, is on the site.

